

**Department of Computer Science and
Engineering(CSE)**
INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG(IIUC)

**Course code:CSE-3524 Course
Title:Microprocessor & Microcontroller Sessional**

EXPERIMENT 2

Experiment Name: Familiarization with PUSH POP Operation, loading the program code and verification of program with it.

OBJECTIVES:

1. To write a program for exchanging numbers using PUSH/POP.
2. To generate the list file.
3. To observe the equipments of kit box.
4. To check the register values in kit box.
5. To analyze the result.

NECESSARY EQUIPMENT:

1. Emu8086 software to run the program.
2. Notepad to write the program.
3. 8086 microprocessor kit.

PROBLEM STATEMENT:

Write an assembly language program to load 16 bit Hex number in two general purpose registers. Interchange their contents using PUSH-POP instruction. Take the initial address, ,SP= 2009H.

Exp2: Interchanging the contents of two register using PUSH-POP;

```
.MODEL SMALL .CODE  
  
MOV SP, 2009H MOV AX, 5566H MOV BX, 7788H  
PUSH AX  
PUSH BX POP  
AX  
POP BX  
END
```

DATA TABLE:-

Instruction	AX	BX	CX	DX	SP	Set Flag
MOV SP, 2009H						
MOV AX, 5566H						
MOV BX, 7788H						
PUSH AX						
PUSH BX						
POP AX						
POP BX						

REPORT WRITING:-

- 1. Include experiment name, objectives, necessary software/ equipment and problem statement.
- 2. Insert the printed copy of “list file” of the program.
- 3. Draw the relevant memory maps.
- 4. Find the result of all related registers from kit box.
- 5. Write discussion if you have any error in the program or troubleshooting of the program.
- 6. Write the answer of the related questions given below.

RELATED QUESTIONS:-

- 1. Can you interchange the contents of two registers with another way?
- 2. If the answer of question1 is “yes” then write at least two new ways (Assembly Language Program) for interchanging two register contents.

List file:

EMU8086 GENERATED LISTING. MACHINE CODE <- SOURCE.

noname.bin_ -- emu8086 assembler version: 4.08

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=====		
=====		
[LINE]	LOC: MACHINE CODE	SOURCE
=====		
=====		

[1]	:	
[2]	0000: BC 09 20	MOV SP, 2009H;
[3]	0003: B8 66 55	MOV AX, 5566H;
[4]	0006: BB 88 77	MOV BX, 7788H;
[5]	0009: 50	PUSH AX ;
[6]	000A: 53	PUSH BX;
[7]	000B: 58	POP AX ;

[8] 000C: 5B
[9] :

POP BX ;
END